

PROJECT FINAL REPORT

An Interactive Tableau Dashboard for Business Intelligence

Food Ordering Behaviour and Consumer Trends

Turning 50,000+ raw food-delivery orders into role-specific insight for Analysts, Operations Managers and Customers

Field	Detail
Team ID	SWTID-2026-5277
Project Name	Food Ordering Behaviour and Consumer Trends
Team Leader	Rishika Bhati
Team Members	Reet Lakhera, Yogita Sain, Vinita Sharma, Chhavi Sharma
Tool / Platform	Tableau Desktop & Tableau Cloud
Report Date	15 July 2026

1. Introduction

1.1 Project Overview

The food delivery industry has become intensely competitive, and platforms now sit on mountains of order data without an easy way to turn it into decisions. This project — Food Ordering Behaviour and Consumer Trends — takes a real 50,000-row food-ordering dataset (order value, delivery fee, cuisine, meal type, city, time taken to deliver, ratings, mood, hunger level, company, and rainy-weather flags across 19 fields) and turns it into an interactive, role-aware Tableau dashboard.

Rather than building one generic report, the team designed the solution around three real personas who each touch the data differently: Rahul Sharma, a Business Analyst who needs to spot order and cuisine trends; Priya Mehta, an Operations Manager who needs to catch delivery delays and peak-hour workload before they become complaints; and Neha Reddy, an everyday customer whose loyalty depends entirely on getting the right food, fast, at a fair price. Every canvas in this report — the empathy map, the journey map, the architecture — is built around keeping these three people, not just the data, at the centre of the design.

1.2 Purpose

The purpose of this project is to design and publish a Tableau-powered analytics solution that:

- Converts a large, multi-dimensional, flat CSV file into structured, drill-down dashboards without requiring end users to write code or SQL.
- Surfaces city-wise demand, high-performing cuisines, and peak meal-time patterns for business decision-making.
- Flags delivery-time bottlenecks and peak-hour workload imbalances so operations teams can rebalance staffing before service quality drops.
- Closes the loop back to the customer experience — faster, more reliable delivery and better cuisine fit is the ultimate measure of success.

In short, the purpose is to replace scattered CSV exports, manual pivot tables and ad-hoc phone coordination with one shared, always-current, self-serve analytics layer that every stakeholder can trust.

2. Ideation Phase

2.1 Problem Statement — Customer Problem Statement

The team framed the problem using an “I am / I'm trying to / But / Because / Which makes me feel” template across the three personas, to make sure the solution was grounded in real frustration rather than an assumed feature list.

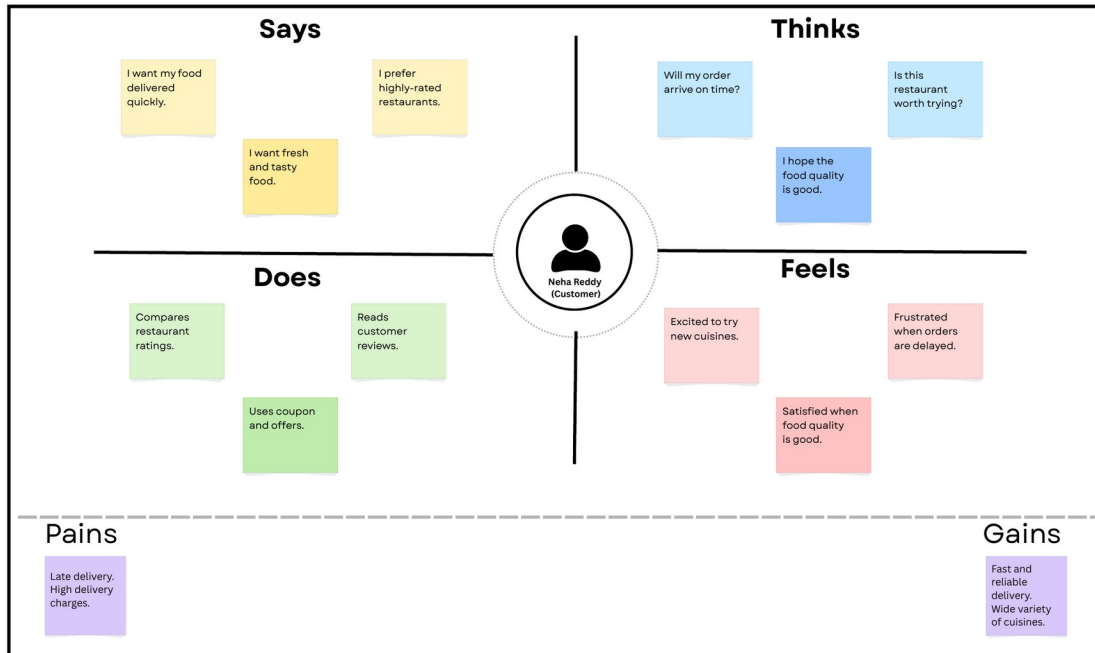
#	I am	I'm trying to	But	Because	Feeling
PS-1	Rahul Sharma (Business Analyst)	Identify high revenue trends and top cuisines	I can't see which segments spend the most	The data is stuck in a flat CSV file	Frustrated and blind to market trends
PS-2	Priya Mehta (Operations Manager)	Monitor delivery costs and peak order times	I don't know when or where bottlenecks happen	We lack a geographic tracking dashboard	Stressed by operational inefficiencies
PS-3	Neha Reddy (Customer)	Order good quality food with quick delivery	I'm not always sure which restaurants provide the best experience	There is limited information about delivery performance and customer satisfaction	Unsatisfied when expectations are not met

After brainstorming and prioritisation (Section 2.3), the team selected PS-2 (Priya Mehta – Operations Manager) as the anchor problem statement, since delivery-performance and peak-hour analysis covers the widest range of business challenges, is the most feasible to build in Tableau with the available fields, and still generates insight that benefits Rahul and Neha's problems as a by-product.

2.2 Empathy Map Canvas — Develop Shared Understanding & Empathy

To make sure the dashboard would actually be used and not just admired, the team built an empathy map for Neha Reddy, the end customer, so that operational metrics could always be traced back to a real human waiting for dinner.

Empathy Map



Empathy Map — Neha Reddy (Customer). Source: empathy_map.pdf

Pains and Gains

The empathy exercise distilled Neha's experience into two clear buckets that directly shaped the dashboard's Operations-Manager view:

Pains	Gains
Late delivery — the single biggest source of frustration and lost loyalty.	Fast, reliable delivery — the biggest driver of satisfaction and reorder rate.
High / unclear delivery charges that feel like a hidden add-on at checkout.	Wide variety of cuisines that match mood, budget tier and craving.
Uncertainty about whether a restaurant is “worth trying” before ordering.	Confidence from ratings and reviews that set correct expectations.

Says / Thinks / Does / Feels in brief: Neha says she wants her food delivered quickly and prefers highly-rated restaurants; she thinks about whether the order will arrive on time and whether the food quality will be good; she does compare ratings, read reviews and use coupons; and she feels excited to try new cuisines but frustrated whenever an order is delayed. This is exactly the tension the Operations Manager dashboard (Section 3.2, FR-2) is built to resolve — late delivery is a pain for the customer and a KPI for the business at the same time.

2.3 Brainstorming

Step 1 — Team gathering & problem statement selection: the team weighed all three candidate problem statements against project goals and the available dataset.

Brainstorm & Idea Prioritization

Team gathering & Problem Statement Selection

The team conducted a brainstorming session to understand the project objectives and evaluate all the problem statements. After evaluating each scenario, the team selected the most suitable problem statement based on the project goals and dataset.

Step-1

Selected Problem statement

PS-2 Priya Mehta(Operational Manager)	Why?
Focus : • Delivery Performance • Peak Orders Hours • City Wise Demand • Operational Efficiency	• Covers multiple business challenges • Helps optimize delivery operations • Easy to analyze using Tableau • Generates meaningful business insights • Matches the available data set.

The team reviewed all the problems statements and finalized PS-2 as it aligns best with the project goals and available dataset for meaningful analysis in Tableau.

Selected Problem Statement — PS-2, Priya Mehta (Operations Manager). Focus: delivery performance, peak order hours, city-wise demand, operational efficiency.

Step 2 — Open brainstorm: five team members individually generated ideas in response to the prompt “How might we improve food delivery performance and customer satisfaction?”

Step 2

Brainstorm

How might we improve food delivery performance and customer satisfaction?

Person 1	Person 2	Person 3	Person 4	Person 5
Analyze peak order hours to optimize delivery staff allocation.	Identify high-demand cuisines to improve inventory planning.	Analyze delivery delays and identify bottlenecks.	Compare city-wise demand for better resource allocation.	Create an interactive Tableau dashboard for decision-making.
Identify weekdays and weekends with the highest order volume.	Compare cuisine popularity across different cities.	Measure average delivery time across different cities.	Analyze order distribution across different locations.	Build KPI cards for Total Orders, Revenue and Average Delivery Time.
Study meal-time demand (Breakfast, Lunch, Dinner) to improve delivery planning.	Analyze customer preferences for different meal types.	Identify factors affecting delivery efficiency.	Identify top-performing cities based on total orders.	Add filters for City, Cuisine and Meal Type to improve dashboard usability.

Individual brainstorm output across the team, grouped by contributor.

Step 3 — Grouping & clustering: overlapping ideas were consolidated into three focus areas that became the backbone of the dashboard's information architecture.

Grouping & Clustering of Brainstorming Ideas

The team organized similar brainstorming ideas into logical clusters to simplify analysis and identify the major areas of project.

Time & Demand Analysis

- Peak order hours
- Meal-time trends
- Weekday vs Weekend analysis
- Cuisine & meal preferences

Geo-Spatial Analysis

- City-wise demand
- Order distribution
- Top-performing cities
- Location-based cuisine trends

Efficiency & Dashboard

- Delivery delay ideas
- KPI cards
- Interactive filters
- Dashboard design

The team grouped all brainstorming ideas into three major categories : Time & Demand Analysis, Geo-Spatial Analysis, and Dashboard & Operational Efficiency. This helped identify the key focus area for dashboard development business analysis.

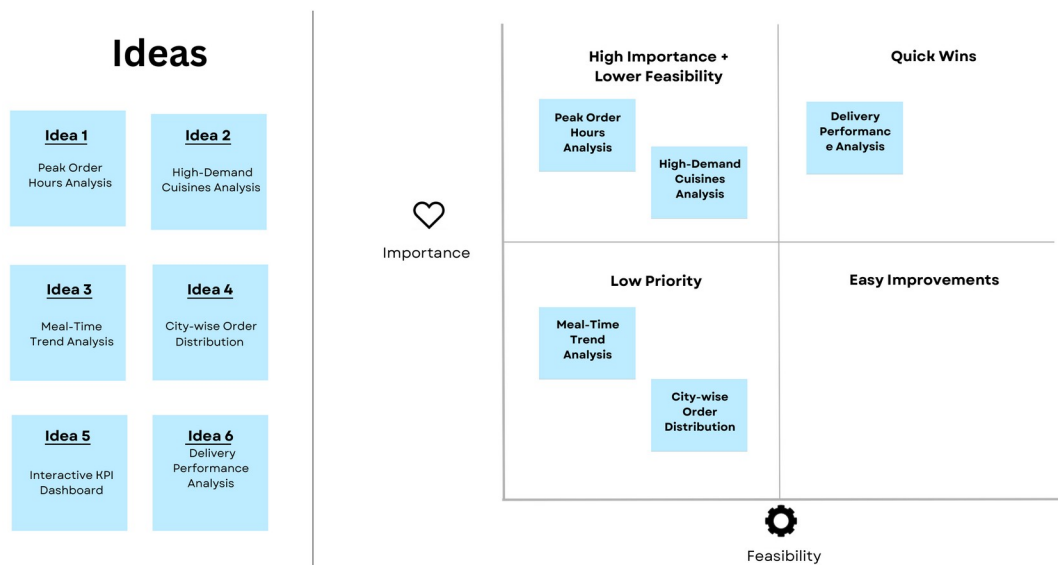
Ideas clustered into Time & Demand Analysis, Geo-Spatial Analysis, and Efficiency & Dashboard.

Idea Prioritisation — Importance vs Feasibility

The clustered ideas were then plotted on an Importance vs Feasibility matrix to decide build order.

Prioritizing Ideas (Importance vs Feasibility Matrix)

The team evaluated all brainstorming ideas based on their business impact and implementation feasibility to identify the most valuable features.

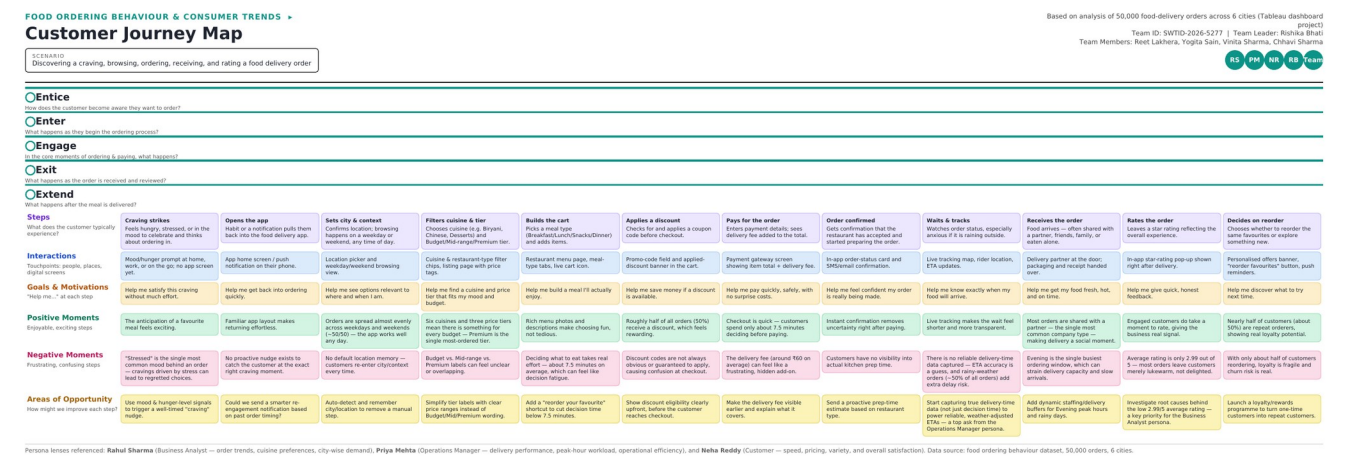


Prioritizing Ideas matrix. Delivery Performance Analysis landed in Quick Wins (high importance, high feasibility); Peak Order Hours and High-Demand Cuisine Analysis were High Importance but lower feasibility; Meal-Time Trend and City-wise Order Distribution were treated as lower priority for the first release.

3. Requirement Analysis

3.1 Customer Journey Map

The journey map traces Neha Reddy's experience across five stages — Entice, Enter, Engage, Exit, Extend — from the moment a craving strikes to the decision to reorder. It was built directly from dataset patterns (mood, hunger level, discount usage, company type, rainy-weather flag) rather than guesswork, so every “negative moment” row maps to a real, measurable field.



Customer Journey Map — Food Ordering Behaviour & Consumer Trends (based on 50,000 orders across 6 cities).

Two findings from the journey map materially shaped the requirements below: (1) “Stressed” is the single most common mood behind an order, and 50% of all orders occur in rainy weather, so delivery-time reliability is not a nice-to-have, it is the top pain point; and (2) with an average rating of only 2.99 out of 5, and only about half of customers reordering, the platform is retaining loyalty by a thread — which is why the Business Analyst dashboard treats repeat-order rate as a first-class KPI.

3.2 Solution Requirement

Functional Requirements

FR No.	Requirement (Epic)	Key Sub-Requirements
FR-1	Order Trend & Cuisine Analysis (Rahul Sharma, Business Analyst)	Order volume by cuisine, meal type, city; order value & delivery fee distribution by segment; demand trend tracking over time.
FR-2	Delivery Performance & Operational Efficiency (Priya Mehta, Operations Manager)	Delivery-time distribution by city; peak ordering hours & workload heat-map; delay/inefficiency flagging.
FR-3	Customer Ordering Behaviour & Satisfaction Tracking (Neha Reddy, Customer)	Cuisine/meal/restaurant-type preference tracking; correlate price & delivery time with rating; monitor repeat-order rate.
FR-4	Interactive Dashboard & Reporting	KPI cards (top cities, high-demand cuisines, peak meal times, delivery-time distribution); city/cuisine/meal/day-type filters; drill-down to order-level detail.

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Dashboards present trends and delivery metrics through clear, intuitive visuals so non-technical stakeholders can interpret insights without training.
NFR-2	Security	Customer order data (user ID, order value, location) is stored and accessed securely, restricted to authorised team members.
NFR-3	Reliability	The dashboard consistently reflects accurate, up-to-date order and delivery data.
NFR-4	Performance	Dashboards load and render quickly even across 50,000+ order records.
NFR-5	Availability	The solution is accessible whenever stakeholders need to review trends or performance, with minimal downtime.
NFR-6	Scalability	The solution handles growing order, city and cuisine volumes without degrading dashboard performance.

3.3 Data Flow Diagram

Two levels of DFD map how raw order data moves from customers and delivery partners, through preprocessing and Tableau analytics, back out to role-specific dashboards.

Figure 1: DFD Level 0 (Context Diagram) — Food Ordering Behaviour & Consumer Trends

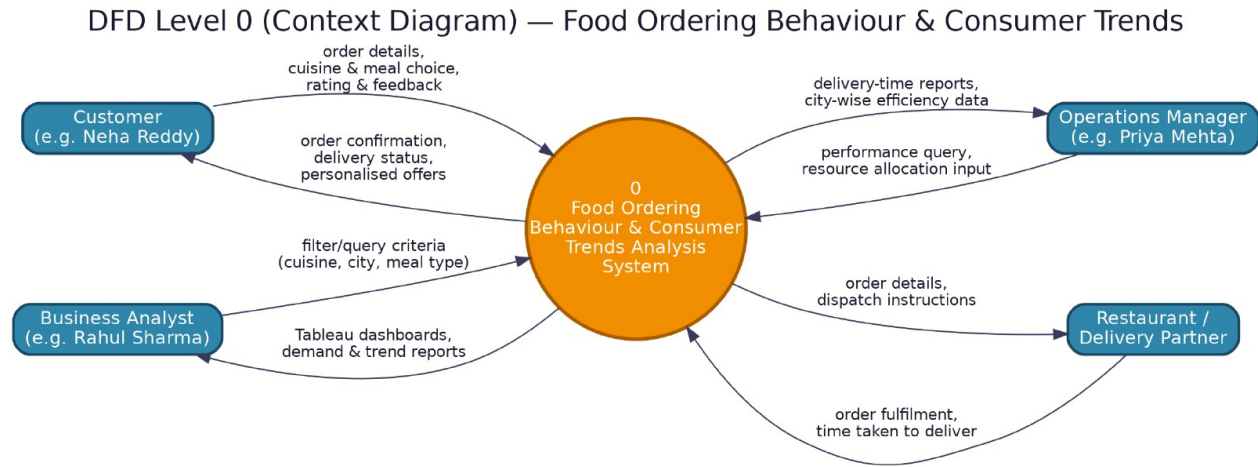


Figure: DFD Level 0 (Context Diagram) — Food Ordering Behaviour & Consumer Trends.

Figure 2: DFD Level 1 — Food Ordering Behaviour & Consumer Trends Analysis System

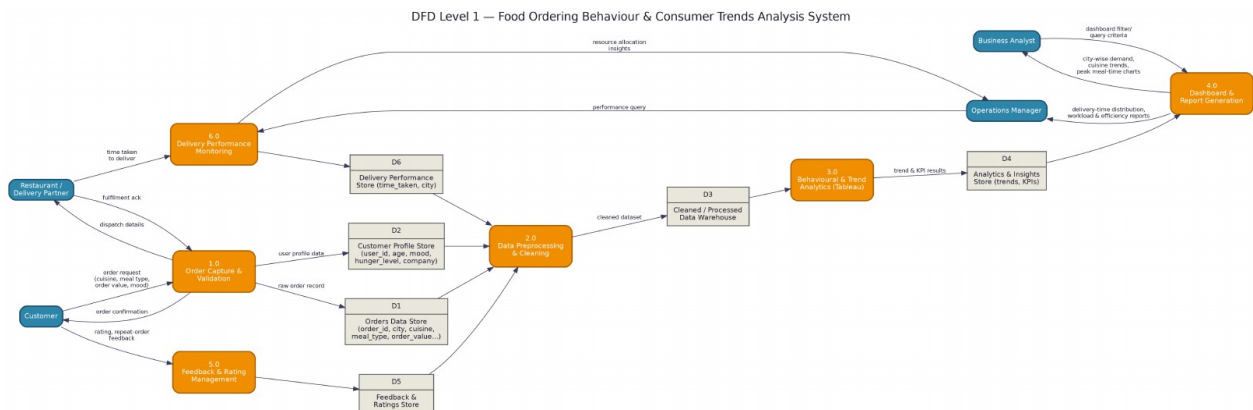


Figure: DFD Level 1 — decomposed into Order Capture & Validation, Data Preprocessing & Cleaning, Behavioural & Trend Analytics (Tableau), Dashboard & Report Generation, Feedback & Rating Management, and Delivery Performance Monitoring, with six data stores (D1–D6).

3.4 Technology Stack

A single Tableau-centred pipeline serves all three stakeholder scenarios — Order Trend & Cuisine Analysis, Delivery Performance Monitoring, and Customer Ordering Behaviour & Satisfaction Tracking.

Project Design Phase-II

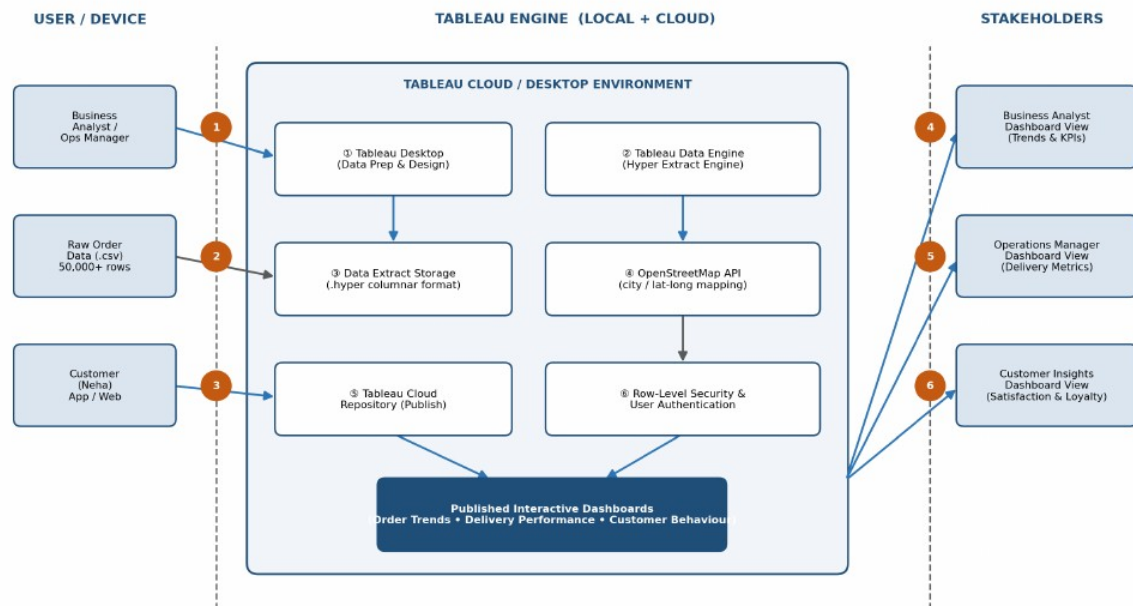
Technology Stack (Architecture & Stack)

Date	11 July 2026
Team ID	SWTID-2026-5277
Project Name	Food Ordering Behaviour and Consumer Trends
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows the end-to-end technical architecture used for this project, covering the three usage scenarios: **Rahul Sharma (Business Analyst)**, **Priya Mehta (Operations Manager)**, and **Neha Reddy (Customer)**. Raw order data flows from the local CSV source into the Tableau processing engine, where it is prepared, mapped, extracted and published as interactive dashboards for each stakeholder.

Scenario coverage: Order Trend & Cuisine Analysis, Delivery Performance Monitoring, and Customer Ordering Behaviour & Satisfaction Tracking, all powered by a single Tableau data pipeline sourced from the food ordering behaviour dataset.



Guidelines followed: application logic blocks shown (Tableau Desktop, Data Engine, Extract Storage), infrastructural demarcation between local (CSV / Desktop) and cloud (Tableau Cloud Repository) components, external interface indicated (OpenStreetMap API), and data storage components (.hyper extract format) clearly labelled.

Technical architecture: local CSV → Tableau Desktop (data prep) → Tableau Data Engine (.hyper extract) → OpenStreetMap API (city mapping) → Tableau Cloud (publish, row-level security) → role-specific published dashboards.

#	Component	Technology
1	User Interface	Tableau Web Dashboard UI / Web Browser

#	Component	Technology
2	Application Logic - 1	Tableau Data Engine (Hyper Technology)
3	Application Logic - 2	Tableau Desktop
4	Database	Flat structured data file (.csv)
5	Cloud Database	Tableau Cloud Server Repository
6	File Storage	Local NTFS / APFS filesystem
7	External API	OpenStreetMap API (lat/long city mapping)
8	Infrastructure	Tableau Cloud (SaaS) / hosted server

4. Project Design

4.1 Problem-Solution Fit

The Problem-Solution Fit canvas connects each persona's jobs-to-be-done, triggers, emotional before/after state, and existing (inadequate) alternatives to the proposed Tableau solution, and names the root cause explicitly: order data is large and multi-dimensional but scattered across raw CSV records with no structured analytics layer, so patterns like peak-hour delays or under-performing cuisines stay invisible.

Canvas Block	Summary
Customer Segments	Rahul (Business Analyst), Priya (Operations Manager), Neha (Customer).
Jobs-to-be-Done	Spot trends fast; catch delivery delays; get quick, fair, varied food.
Triggers	Monthly reviews; rising complaints during rush; hunger, discounts, cravings.
Emotions Before → After	Overwhelmed / frustrated / anxious → confident / in control / satisfied.
Existing Alternatives	Manual Excel pivots, generic BI tools, static reports, phone coordination — slow and non-interactive.
Constraints	Low coding skill in the team; budget limits on BI licensing; 50,000+ rows can slow basic tools; low tolerance for delay.
Root Cause	Multi-dimensional order data with no structured analytics layer — patterns stay invisible.

4.2 Proposed Solution

Parameter	Description
Problem Statement	Businesses cannot easily convert large, multi-dimensional food-order data into decisions on trends, delivery performance and customer experience — there is no single, structured, easy-to-use analytical view connecting Analyst, Operations Manager and Customer needs.
Idea / Solution	An interactive Tableau-based analytics solution that ingests the raw order dataset, processes it through Tableau's Hyper engine, and publishes role-specific dashboards to Tableau Cloud.
Novelty / Uniqueness	Purpose-built around three real stakeholder journeys rather than one generic view; combines rarely-paired behavioural fields (mood, hunger level, company, rainy weather) with operational fields (delivery fee, time taken, repeat orders); uses the free OpenStreetMap API instead of a paid GIS layer.
Social Impact / Satisfaction	Faster detection of delivery delays and bottlenecks improves reliability — the single biggest driver of customer satisfaction and reorder behaviour — and supports fairer pricing and restaurant partnerships.
Business Model	Supports existing revenue streams (delivery fees, commission, priority delivery) by directing marketing and delivery resources to the highest-return city/cuisine/meal-time segments; the dashboard can also be offered as a paid analytics module to restaurant partners.
Scalability	Tableau's columnar .hyper extract format scales from 50,000 to several million rows without redesign; new cities, cuisines or stakeholder roles can be added on top of the same data model.

4.3 Solution Architecture

The architecture maps how each persona's raw input flows through Tableau Desktop, the Hyper data engine, OpenStreetMap city mapping, and Tableau Cloud's row-level security, into one set of published interactive dashboards.

Project Design Phase

Solution Architecture

Date	11 July 2026
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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For the Food Ordering Behaviour and Consumer Trends project, it maps how raw order data from the three usage scenarios (Rahul Sharma – Business Analyst, Priya Mehta – Operations Manager, Neha Reddy – Customer) flows through the Tableau analytics pipeline into published, role-specific dashboards. Its goals are to:

- Find the best tech solution to solve existing business problems – turning raw food-ordering data into decisions on trends, delivery performance, and customer experience.
- Describe the structure, characteristics, behaviour, and other aspects of the software to project stakeholders (Analyst, Operations Manager, Customer).
- Define features, development phases, and solution requirements for the Tableau-based analytics pipeline.
- Provide specifications according to which the solution is defined, managed, and delivered – from raw .csv ingestion through to published Tableau Cloud dashboards.

Solution Architecture Diagram:

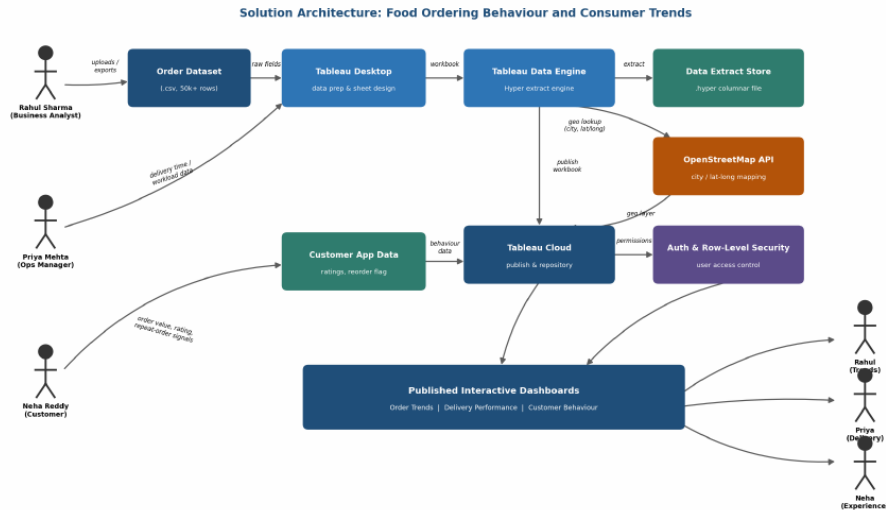


Figure 1: Architecture and data flow of the Food Ordering Behaviour and Consumer Trends analytics solution — from raw order data and customer/operations inputs, through Tableau Desktop, the Hyper data engine, the OpenStreetMap API for city mapping, Tableau Cloud publishing with row-level security, to the published dashboards consumed by each stakeholder.

Reference: https://help.tableau.com/current/pro/desktop/en-us/hyper_data_source.htm
<https://www.openstreetmap.org/about>

Solution Architecture — Food Ordering Behaviour and Consumer Trends. Reference: help.tableau.com (Hyper data source), [openstreetmap.org/about](https://www.openstreetmap.org/about).

5. Project Planning & Scheduling

5.1 Project Planning

The 14 user stories defined during requirement analysis (Section 3.2) were split across two one-week sprints, sequencing the highest-priority, highest-feasibility items (order capture, trend dashboards, delivery-time distribution, peak-hour heat-maps, data validation) into Sprint 1, and satisfaction-tracking, comparative and access-control features into Sprint 2.

Sprint	Functional Requirement	User Stories	Story Points	Priority	Sprint Duration
Sprint-1	Order Placement & Tracking (USN-1, USN-3)	2	13	High / Medium	06 Jul – 10 Jul 2026
Sprint-1	Order & Demand Trend Analysis (USN-6, USN-7)	2	16	High	06 Jul – 10 Jul 2026
Sprint-1	Delivery Performance & Resource Allocation (USN-10, USN-11)	2	16	High	06 Jul – 10 Jul 2026
Sprint-1	Order Placement Pricing & Data Management (USN-2, USN-13)	2	16	High	06 Jul – 10 Jul 2026
Sprint-2	Feedback & Rating (USN-4, USN-5)	2	13	High / Medium	11 Jul – 15 Jul 2026
Sprint-2	Spending Pattern & Demand Tracking (USN-8, USN-9)	2	10	Medium	11 Jul – 15 Jul 2026
Sprint-2	Weather Impact & Access Control (USN-12, USN-14)	2	8	Medium / Low	11 Jul – 15 Jul 2026

Sprint Duration and Velocity

Sprint	Total Story Points	Duration (Days)	Velocity (Points / Day)	Release Date
Sprint-1	61	5	12.2	10 July 2026
Sprint-2	31	5	6.2	15 July 2026

Sprint 1 was intentionally front-loaded with the core data pipeline, trend dashboards and delivery-performance views, which is why its velocity (12.2 points/day) is roughly double Sprint 2's (6.2 points/day) — Sprint 2 focused on lighter comparative views, feedback capture and access-control, which required less new pipeline work and mostly reused the Sprint-1 data model.

6. Functional and Performance Testing

6.1 Performance Testing

Testing was carried out directly against the published Tableau dashboard, connected to food_ordering_behavior_dataset.csv as a Text File data source.

Parameter	Result
Data Rendered	50,000 records rendered across 19 fields with 0 null values. Tableau auto-classified 13 Dimensions and 6 Measures. Row-count in the Data Source pane matched the source CSV exactly, confirming a clean, lossless render.
Data Preprocessing	Fields renamed for readability (e.g. time_taken_to_order → Prep Time (mins)); Discount Applied, Is Repeat Order and Rainy Weather converted to Boolean flags; an Age Group bin (18-25, 26-35, 36-45, 46+) created; all 50,000 Order IDs confirmed unique.
Utilisation of Filters	Quick Filters on City (6), Cuisine (6), Meal Type (4), Day Type (2) and Order Time (4); a Context Filter on City to speed up calculation on the 50,000-row extract; a Parameter-based minimum-rating filter; Filter Actions linking the city map to every detail view.
Calculated Fields	Total Order Revenue, Repeat Order Rate, Avg Prep Time, Discount Impact, Age Group bucket, and a Rainy-Day Order Flag — these power the revenue-by-meal-type and age/segment visualisations.
Dashboard Design	9 visualisations combined into one interactive dashboard with tooltips, highlight actions and the filter set above (full detail in Section 7).
Story Design	3 story points, one per persona — Rahul (order trends & cuisine performance), Priya (delivery/prep-time distribution & peak-hour workload), Neha (satisfaction drivers) — each with caption text and navigate buttons tied to the dashboard.

Key Performance Matrix

Total Orders	Total Transaction Amount	Avg. Rating Given	Avg. Time Taken to Order
50,000	₹30.37M	3	7.50 minutes

7. Results

7.1 Output Screenshots

The published dashboard contains nine linked visualisations. Screenshots below are exported evidence from the live Tableau workbook.

Dashboard Screenshots (Evidence)

The following screenshots, exported from the published Tableau dashboard, provide visual evidence for the values and visualizations summarised in Parameter 5 (Dashboard design) above.

Key Performance Matrix			
Total Orders	Total Transaction Amount	Avg. Rating Given	Avg. Time Taken To Order
50,000	30.37M	3	7.50Min.

Fig. 1 — Key Performance Matrix: Total Orders, Total Transaction Amount, Avg. Rating Given, Avg. Time Taken to Order.

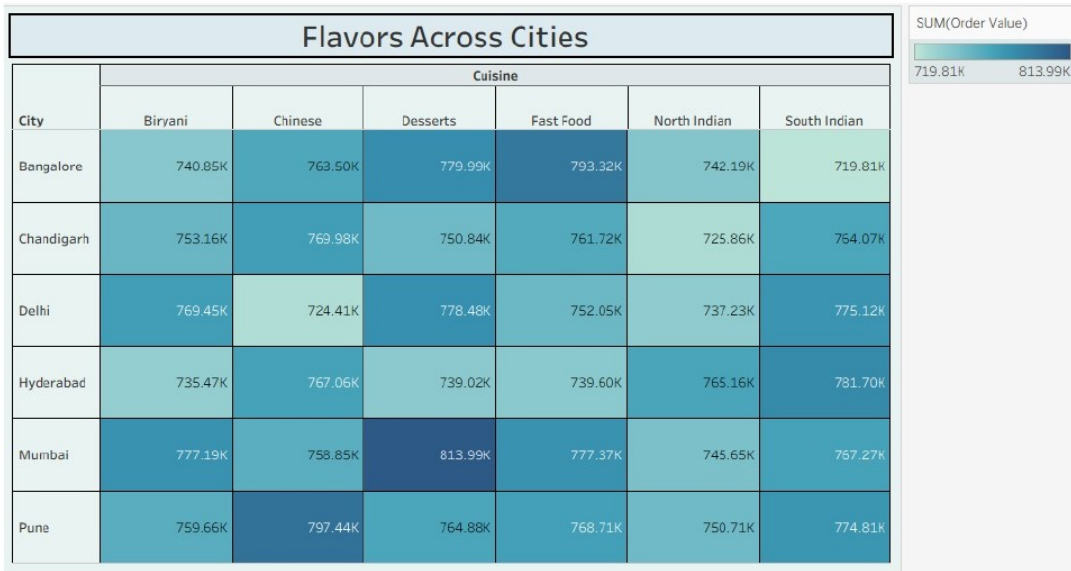


Fig. 2 — Flavors Across Cities: SUM(Order Value) by City and Cuisine.

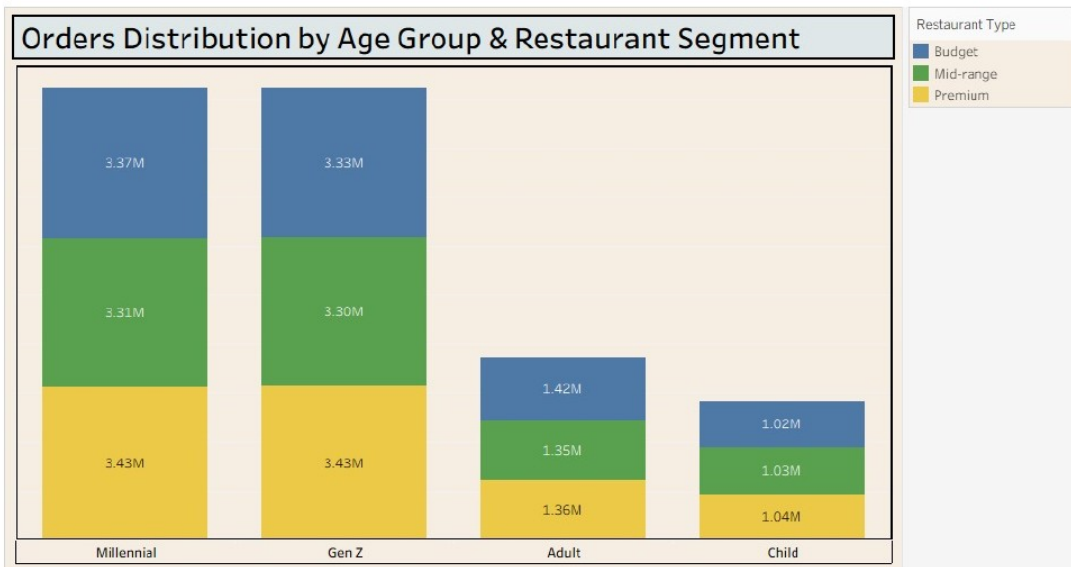


Fig. 3 — Orders Distribution by Age Group & Restaurant Segment (Budget / Mid-range / Premium).

Fig. 1-3 — Key Performance Matrix; Flavors Across Cities (order value heat-map by city × cuisine, Rs 719.81K–813.99K); Orders Distribution by Age Group & Restaurant Segment.

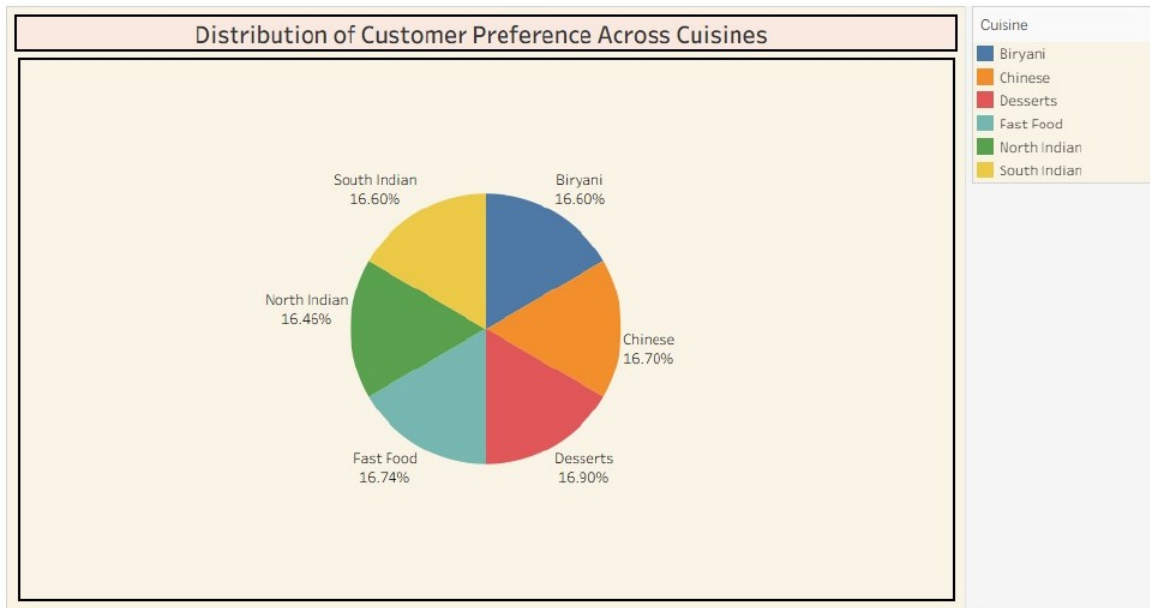


Fig. 4 — Distribution of Customer Preference Across Cuisines.

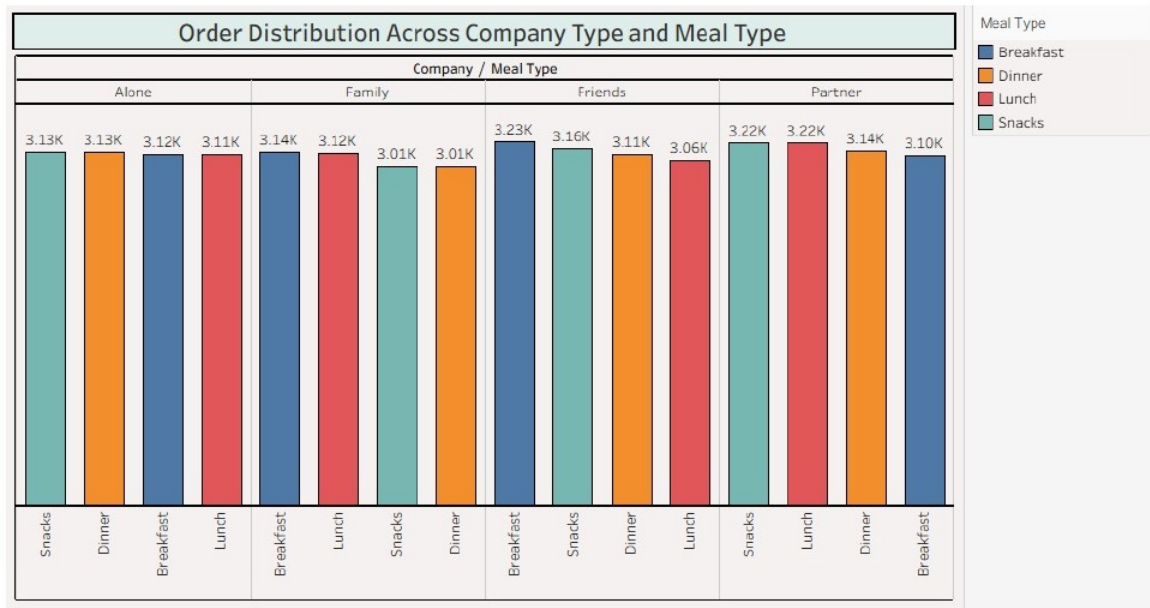


Fig. 5 — Order Distribution Across Company Type and Meal Type.

Fig. 4-5 — Distribution of Customer Preference Across Cuisines (Desserts leads at 16.90%, North Indian lowest at 16.46%); Order Distribution Across Company Type & Meal Type (near-even ~3.0K–3.23K orders per combination).

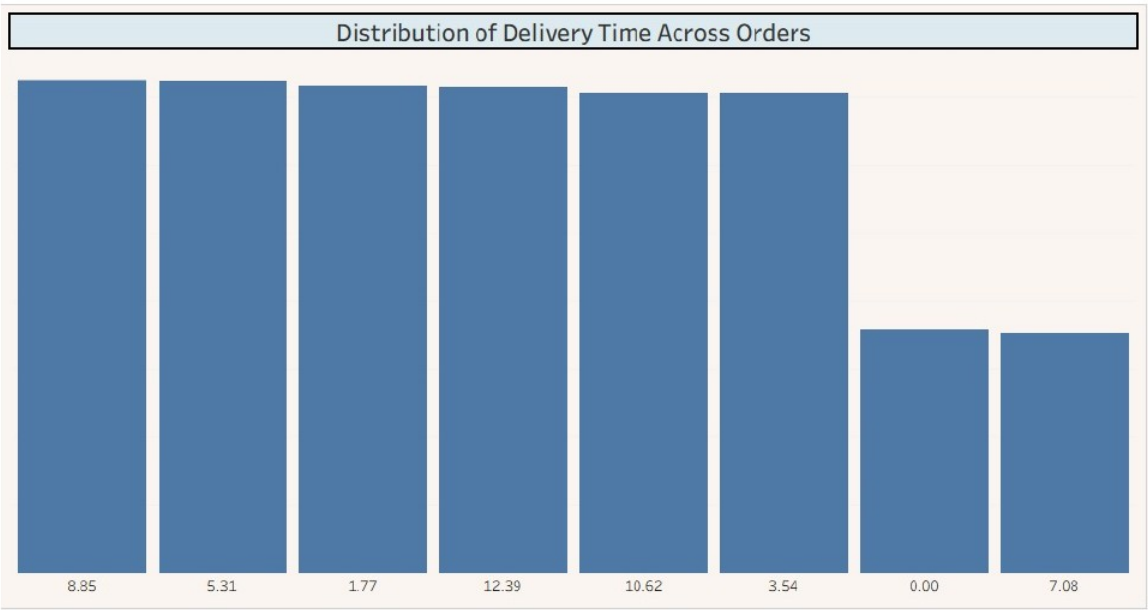


Fig. 6 — Distribution of Delivery Time Across Orders.

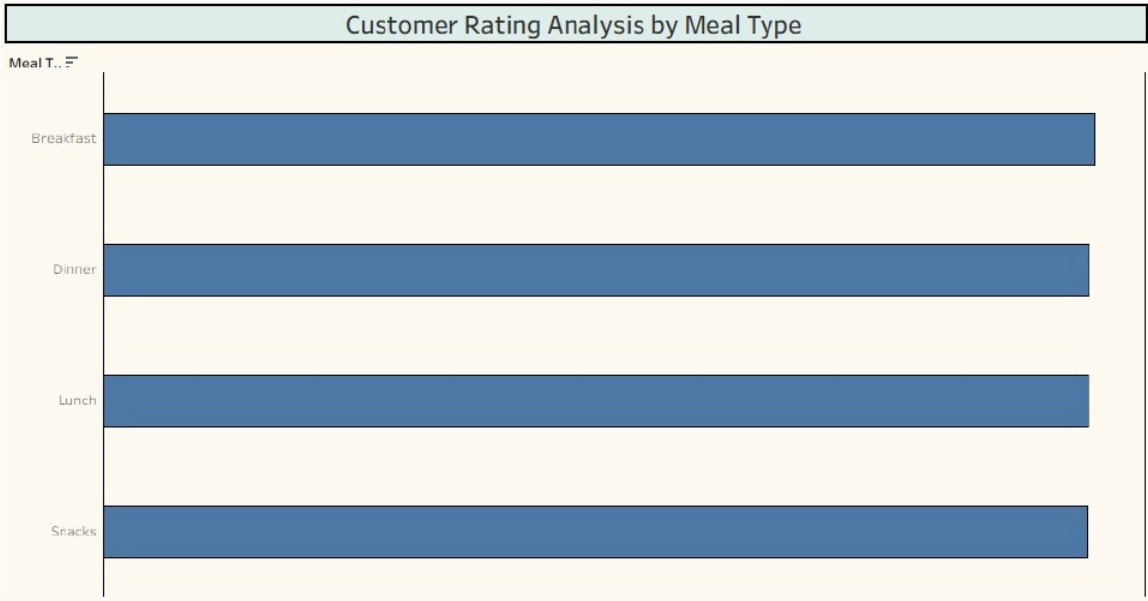


Fig. 7 — Customer Rating Analysis by Meal Type.

Fig. 6-7 — Distribution of Delivery Time Across Orders; Customer Rating Analysis by Meal Type (Breakfast, Lunch, Dinner and Snacks receive comparable average ratings).

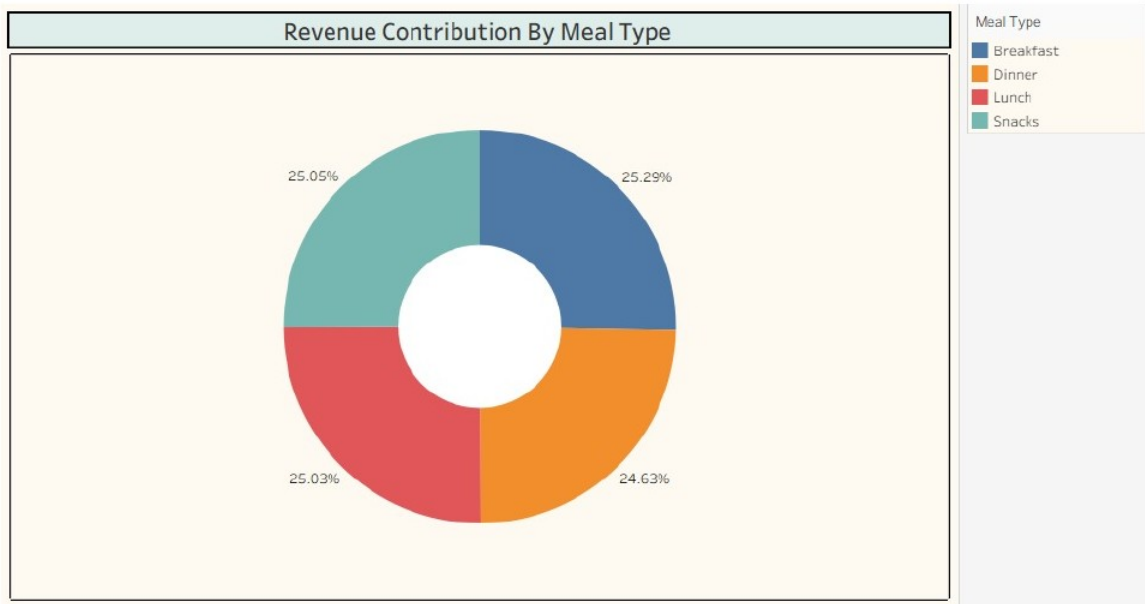


Fig. 8 — Revenue Contribution By Meal Type.

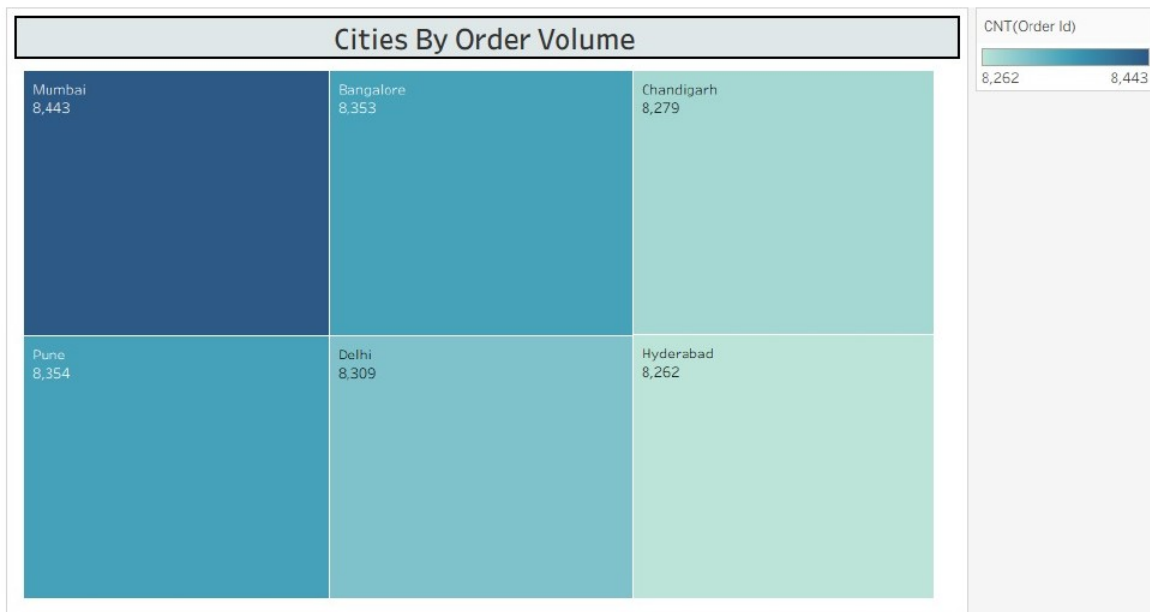


Fig. 9 — Cities By Order Volume.

Fig. 8-9 — Revenue Contribution by Meal Type (an almost even 24.6-25.3% split across Breakfast/Lunch/Dinner/Snacks); Cities By Order Volume (Mumbai highest at 8,443 orders, Hyderabad lowest at 8,262).

Taken together, the results confirm the hypothesis behind PS-2: demand is not concentrated in one city, cuisine or meal time — it is broad and evenly spread — so the real lever for improving customer satisfaction is operational (delivery-time consistency and peak-hour staffing) rather than chasing a single “winning” segment.

8. Advantages & Disadvantages

Advantages

- No-code, visual analysis — Business Analysts and Operations Managers get trend and delay insights without writing SQL or Python.
- Role-specific dashboards mean each stakeholder sees only the metrics relevant to their job, reducing noise and decision fatigue.
- Tableau's Hyper columnar extract scales from 50,000 to millions of rows without a redesign.
- Free OpenStreetMap integration avoids the cost of a paid GIS layer for city-level mapping.
- Filter Actions and drill-down let users move from a summary chart straight to order-level detail in one click.
- Directly closes the loop between operational metrics (delivery time, peak hours) and customer-facing outcomes (rating, repeat-order rate).

Disadvantages

- Tableau Cloud licensing cost can be a constraint for smaller teams, as noted as a customer constraint in the Problem-Solution Fit canvas.
- The current pipeline is batch-oriented (CSV upload) rather than real-time streaming, so dashboards reflect the last refresh, not live order status.
- Row-level security and access control (USN-14) were scoped as Low priority in Sprint 2, so fine-grained permission management is still an early-stage feature.
- Average customer rating in the current dataset is only 2.99/5 — the dashboard surfaces this clearly, but resolving the underlying service issues requires operational action outside the tool itself.

9. Conclusion

This project set out to answer one question: can a structured, role-aware Tableau solution turn 50,000+ rows of raw food-ordering data into insight that a Business Analyst, an Operations Manager and an everyday customer can each act on? The resulting dashboard — nine linked visualisations, city and cuisine heat-maps, delivery-time distributions, and persona-driven story points — shows that it can. Order demand is broad and evenly distributed across cities, cuisines and meal times, which means the biggest satisfaction lever available to the business is operational consistency — catching delivery delays and peak-hour bottlenecks before they affect a customer like Neha — rather than betting on any single high-performing segment.

10. Future Scope

- Move from batch CSV refresh to a near-real-time data connection so Operations Managers can react to delays as they happen, not after the fact.
- Extend row-level security and role-based access control (USN-14) into a fully governed permission model for Analyst vs Operations Manager vs Restaurant Partner views.
- Add a predictive layer (e.g. forecasting peak-hour order volume or delivery-time risk on rainy days) on top of the existing Hyper extract.
- Publish a restaurant-partner-facing view as a paid analytics module, as outlined in the Business Model section of the Proposed Solution.
- Investigate root causes behind the current 2.99/5 average rating as a dedicated follow-up analysis, as flagged in the Customer Journey Map's Areas of Opportunity.

11. Appendix

Source Code

This phase of the project is delivered as a Tableau workbook (data-preparation steps, calculated fields and dashboard layout) rather than a traditional source-code repository; calculated-field logic is documented in Section 6.1.

Dataset

food_ordering_behavior_dataset.csv — 50,000 records × 19 fields (order_id, user_id, age, city, order_time, day_type, cuisine, meal_type, restaurant_type, order_value, discount_applied, delivery_fee, time_taken_to_order, rating_given, is_repeat_order, mood, hunger_level, company, rainy_weather). Verified with zero missing values and zero duplicate records across all columns (Data Quality Report, Section 3).

Team Details

Role	Name
Team ID	SWTID-2026-5277
Team Leader	Rishika Bhati
Team Member	Reet Lakhera
Team Member	Yogita Sain
Team Member	Vinita Sharma
Team Member	Chhavi Sharma